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CSA0811 DATA STRUCTURES

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ACTIVITY - 4 $\Rightarrow$ HASHING PROBLEM

1. Given Input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 } and a hash function $h(x) = x \pmod{10}$, Show the resulting:
- (a) Separate chaining hash table
 - (b) Open addressing hash table using linear probing
 - (c) Using Quadratic probing
 - (d) Second hash function, $h_2(x) = 7 - (x \pmod{7})$

~~Key~~ Given,

keys = 4371, 1323, 6173, 4199, 4344, 9679,

1989, $h(x) = x \pmod{10}$

(a) Separate chain (Hash table)

0	1	2	3	4	5	6	7	8	9
	4371		1323	4344					4199
			6173						9679
									1989

(b) Open Addressing - Linear probing

$$h(\text{key}) = h(\text{key} + 1) \bmod 10$$

i) $4371 \bmod 10 = 1$

ii) $1323 \bmod 10 = 3$

iii) $6173 \bmod 10 = 3 + 1 \Rightarrow 4$

iv) $4344 \bmod 10 = 4 + 1$
 $= 5$

0	1	2	3	4	5	6	7	8	9
9679	4371	1989	1323	6173	4344	-	-	-	4199

(c) Open Addressing - Quadratic probing

$$h(\text{key}) = h(\text{key} + i^2) \bmod 10$$

0	1	2	3	4	5	6	7	8	9
9679	4371	-	1323	6173	4344	-	-	1989	4199

(d) Double hashing,

$$h_1(\text{key}) = x \bmod 10$$

$$h_2(\text{key}) = 7 - (\text{key} \bmod 7)$$

Formula,

$$h_i(\text{key}) = (h_1(\text{key}) + i h_2(\text{key})) \bmod 10$$

Key	$x \bmod 10$	$x \bmod 7$	$\cancel{x \bmod 7}$ $h_2(x) = 7 - (x \bmod 7)$
4371	1	0	7
1323	3	6	1
6173	3	6	1
4199	9	2	5
4344	4	4	3
9679	9	5	2
1989	9	1	6

0	1	2	3	4	5	6	7	8
-	4371	-	1323	6173	9679	-	4344	-